

MY KIND OF SHOW...

Our international traveller Robin Webster presents a full report on innovations in the computer industry from the Consumer Electronics Show in Chicago.

Who would have expected that at an exhibition best known for its displays of the latest home video films, novel telephones, noisy toys and ear-splitting quadrophonic systems for Porsches, there would also be a rash of significant announcements for the microcomputer industry?

As it turned out, few of the 80,000 or so people who attended the Summer Consumer Electronics Show held in Chicago (June 5-9) were ready for what happened.

Main contenders

Video games company Coleco (best known for its Atari-competitive ColecoVision games system) introduced a colour microcomputer called Adam. Or to be more precise, the company introduced a package consisting of: a Z80A-based system unit with 80k of RAM and an integral data storage device that can hold up to 500k; a keyboard with joystick cursor controller; a letter-quality (daisy-wheel printer); ROM-based word processing; a Basic language pack, called Smart Basic; and a games pack, called Buck Rogers — The Planet of Zoom.

The whole system, as described, will cost about \$600, according to the company. Another Adam version, which simply upgrades a ColecoVision video games machine into an Adam system will sell for around \$400.

Commodore, which has already featured in the cut-price computer market, drew much attention with what was described as 'the largest single software introduction in the history of the company'. About 70 new software products for the Commodore 64 and VIC-20 were either on show or being seriously discussed at the Commodore stand. While many of these were games and educational packages, the company slipped in a few demos of a product called 'Magic Desk'. Although it is still in the prototype stage, Magic Desk has some features in common with (albeit at a much cruder level) the Desktop Manager software available on Apple Computer's Lisa system.

Other Commodore events were the introduction of a portable computer called the Executive 64 which not only has a colour screen and costs \$995, but can be made to run CP/M or PET programs with the addition of suitable options; and the welcome announcement that the company intends to sell Microsoft's Multiplan spreadsheet system for under \$100 on the Commodore 64 computer.

Atari, is also trying to shake loose its image of being 'just a games machine maker' with great gusto. It publicly announced four new computers — called the 600XL, 800XL, 1400XL, and 1450XLD —, a CP/M add-on that works with all Atari systems, the AtariWriter

word processing system priced at \$99.95, a 16k Logo language cartridge, a disk-based colouring and drawing package called Paint, a light-pen, a touch-pad, and many more pieces of hardware and software.

More privately, in a nearby hotel suite, Atari was willing to discuss and display some experimental work it has been doing, such as Atari Artworks (a menu-based, free-hand drawing system which provided similar facilities to Apple's LisaDraw) and an as-yet-unnamed free-hand drawing system which combines an on-screen line/shape/and colour palette with the ability to store 'pictures' on visibly displayed cards or windows. Once a picture or a group of pictures are saved in this way, the cards themselves can be electronically put



The Coleco Vision Family Computer System with memory console, Smart Writer and Keyboard.



The Romox Programming Terminal can store up to 500 different games.



The Tomy Tutor for children has a 16-bit TI 9995 chip.

simple 'stepper-motor' design for its product. Instead of being whizzed around at high speed under the control of fancy electromechanics/software, Coleco's print-wheel is driven to discrete, degree-by-degree movements. This old faithful approach is probably more than adequate for the average home computer. There are suggestions that Coleco may be able to push the print speed up from 10cps to 30 or 40cps; if it does, it will have to pay more attention to the noise the printer makes while it is working.

A more quiet rival is the new Atari 1027 'barrel' printer which operates at 20 characters per second. It is about one-third the size of the Coleco device and is very quiet in operation. Rather than using a print-wheel, this printer uses a set of narrow rollers which have the print characters on their outside faces. The rollers are all rotated at once, and somehow the print mechanism is able to impact the correct character, from the correct roller, onto the paper surface. Surprisingly, the printer does not use ink or carbon ribbons. Instead, it features a spongy ink roller which transfers ink to the character faces. Unfortunately, it is priced at \$349.95 and can only produce one typestyle (Prestige Elite 12).

Commodore range

Commodore's presence at the show was very noticeable, partly because of the company's large stand and shuttle service to the Commodore Clipper — a rented boat moored just a few miles away from the main exhibition hall — but mainly because it had so many new things on offer.

Fuelling the home/personal computer price war, Commodore announced large price reductions on its machines: software trade prices were cut by up to 50 per cent; hardware trade prices were cut by up to 25 per cent.

The portable Executive 64 machine, coming in under the \$1000 mark by just \$5, uses a 6510 CPU (this is program compatible with the 6502) and has Basic stored in ROM. It can run a wide range of programming languages in addition to Basic, including Pascal, Comal, Logo, Pilot and Assembler. The main hardware features are that it weighs 27.6 pounds, has 64k RAM, full upper/lower case display and keyboard, an integral six-inch colour display, and one floppy disk drive with only a 170k storage capacity with the option of adding another drive later. It is very compact, measuring 14½ × 14½ × five inches.

By buying an add-on Z80 cartridge, the Executive 64 can be turned into a CPM machine or, with a PET Emulator, a PET. Also, the Executive 64 can make use of

VIC-20 and Commodore 64 peripherals.

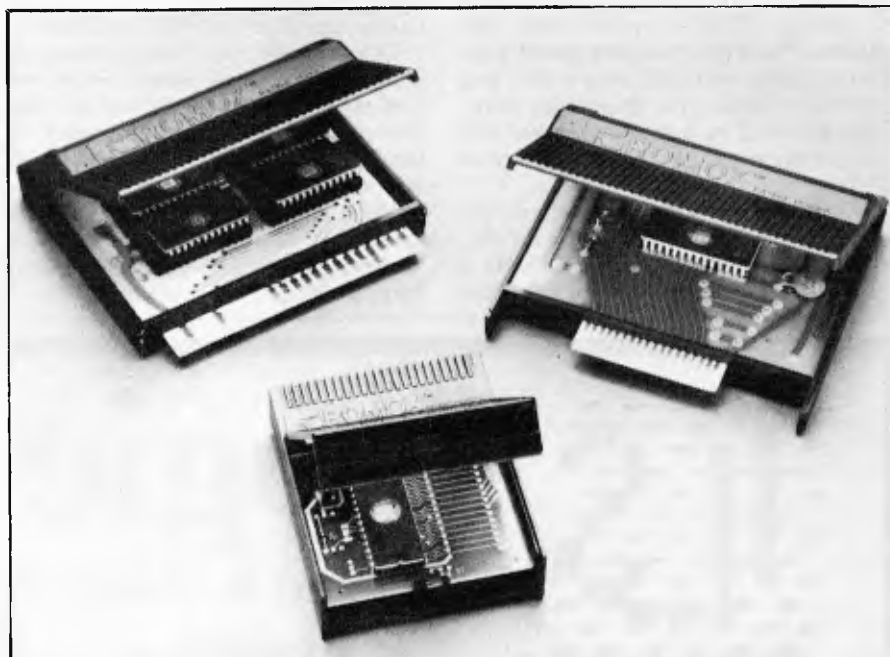
Since Apple's Lisa became widely known, there has been a flurry of announcements regarding similar projects, packages and concepts. Commodore (which has a licence to use the Xerox-developed Smalltalk system), has made a first attempt at producing software that allows a user to enter commands by manipulating visual representations of everyday objects. It is called Magic Desk and runs on the Commodore 64.

With Magic Desk, the user is provided with a graphic representation — in arcade quality resolution — of an office. In the centre of the screen there is a desk with a collection of familiar objects arranged on top: a typewriter (word processing or text editing); a calculator and financial journal (for calculating and simple spreadsheet facilities; a card file (for storing lists of names and addresses); and a telephone

(Commodore wouldn't comment on what the telephone represented since it couldn't be demonstrated, but it doesn't take much imagination to figure that one out). Under the desk is a wastebasket, which is used to discard unwanted files. To the right of the desk is a filing cabinet with a digital clock resting on top of it, and to the left is an artist's easel.

Like the telephone, the artist's easel is not yet available for use, but it will presumably be used to access some kind of drawing system. The filing cabinet, on the other hand, is a key metaphor of the Magic Desk environment since it is where all files will be saved to or retrieved from. The cabinet has three drawers, each of which contains ten file folders. Each of these folders contains ten pages for document storage. It's not clear how much information can be stored per page.

Instead of a conventional cursor, users



Hardware manufacturer Romox made two major announcements at the Consumer Electronics Show. It is going to market a product that gets around Texas Instruments' plans to modify the TI99/4A so that it will accept only TI approved games cartridges. And it will soon launch a juke-box like terminal that will allow retailers to program and reprogram special games cartridges.

The first product is called GamePort. It plugs into the TI99/4A I/O port, and so can communicate directly with the machine's 9900 central processor. In this way, Romox says, users can avoid the consequences of the TI circuit modification. GamePort will play any standard ROM cartridge and it also includes an extra 8k of ROM to expand the TI99/4A's capabilities. There are plans to include some RAM memory as well, but Romox has not yet decided how much should be included. GamePort will cost \$39.95 and will be available later this year.

The company's second product is designed for retail stores that already sell games software cartridges. Instead of stocking a large number of the relevant cartridges, the new terminal allows retailers to maintain a central base of up to 500 games which can be copied any number of times onto what Romox is calling the Edge Connector Programmable Cartridge.

This cartridge is simply inserted in the games terminal and the software is downloaded into it. Once you get tired of a particular game you just take the cartridge back and reload another game over the first (the same way audio cassettes can be used for one recording then erased).

Romox plans to lease the terminal to retail stores for about \$100 per month.

of Magic Desk move a disembodied arm, complete with pointing finger, to select any of the office objects, or metaphors. If you want to open a file cabinet drawer, for example, you would use a joystick, mouse, trak-ball, or any other cursor controller to

first point at the relevant drawer with the hand. You would then press the controller's 'select' or 'fire' button to execute the 'open file-drawer' command, and then move onto finding a particular file and updating it in some way.

Magic Desk does not presently feature windows as on the Xerox Star, Apple Lisa, or even those simple bordered split screens available with Microsoft's Multiplan. Instead, whenever a new activity is selected, the whole screen display is updated. Magic Desk has little of the sophistication of the Star and Lisa, but yet it shows the kind of effort being made at the home computer software market level. It is simple to use, can be immediately understood by a user regardless of his or her native language, and the on-screen metaphors can be customised more or less at will. Magic Desk I, as this first system is being called, will be available later this year in ROM cartridge form for less than \$100. A second cartridge, to handle calculating and home applications, is also on the way.

Atari attack

With the announcement of four new machines (all based on the 6502C chip running at 1.79MHz) and a range of new peripherals, Atari is hoping to quiet much of the criticism it has been receiving in the last year or so.

The smallest of the new entrants, the 600XL, comes with 16k RAM and 24k ROM as standard, although the RAM can be pushed up to 64k with an optional Memory Module expansion unit. A useful addition is the 'Help' key, placed to the



Following the unsuccessful launch of the Aquarius home computer last autumn its maker Mattel Electronics produced a re-vamped model, the Aquarius II, at the show.

It includes full-stroke keyboard, 12k ROM, 64k RAM, a Z80A processor and it runs Microsoft Basic with 16 colours.

Back-up marketing includes the Aquarius COM/PAC package containing the basic unit with hand controllers for game-playing, a 40 column thermal printer and a cassette recorder.

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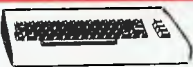


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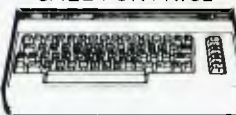
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MY KIND OF SHOW...

right of the keyboard along with other special function keys. If the user encounters some difficulty in proceeding with a task, pressing the Help key will usually summon up useful instructions. This feature is common to all of the new machines, as is the ability to generate 256 colours, to emit all sorts of sounds, and to carry out self-test diagnostics.

In addition, the 800XL has 64k as standard; the 1400XL has 64k RAM, four programmable keys, a built-in modem, and a speech synthesiser. The 1450XLD includes all of this, plus an integral double-sided, dual-density disk drive.

Only the 600XL has been given a price — \$199. Atari has yet to make up its mind about the three others.

As a result of the new Atari CP/M plug-in module, produced by California company, Add-On Software, all Atari computers can now run CP/M version 2.2 programs. The module features a four MHz Z80 micro with 64k RAM and it connects to the various machines via the serial I/O port. While this is a fairly simple

approach, there is a penalty to pay as far as program execution speed is concerned because the Z80 and 6502C chips can only communicate at the standard I/O port speed rather than at system bus speed. No price has been set.

Conclusions

In the midst of all this jockeying for position by the major competitors, it would be easy to miss some interesting advances made by less conspicuous companies. It would be easy to miss the system with a 10.7 MHz chip, called Tomy Tutor, which has been designed for younger children. It comes with a peanut-butter proof (rather than ergonomic) keyboard, and can *really* handle animated graphics. Standard memory is 32k ROM and 16k RAM, expandable up to 64k. Interested parents/children can obtain the machine on a 'five day free home trial' basis.

Just a short time ago, a company called Spectravideo would have made quite a splash with its two microsystems — the

SV-318 and the SV-328. Both machines are based on the Z80A, and can handle up to 256k RAM and 96k ROM. The SV-318 comes with built-in joystick cursor controller, special word processing keys, 32 graphic sprites, ten user definable keys and CP/M compatibility for under \$300. The SV-328 is designed more for business use and, in addition to most of the SV-318 features, comes with Basic, word processing, a 'Super Terminal' program and a 'Help' program stored in ROM. The SV-328 costs \$595.

With Commodore, Coleco and Atari engaged in an all-out battle for the home computer market, we'll obviously have to look that bit harder so as not to miss significant breakthroughs by the smaller companies.

Overheard at the show

Apple's Mackintosh will probably be offered with TK! Solver, the high powerful calculating tool from Software Arts.

The Mackintosh is a low cost personal computer which will be a cut-down version of Lisa running similar applications and incorporating a mouse, although not the icons. It was named after a breed of the edible apple in the US, the McIntosh.

END

Sendata 300 Modem Direct Connect

A new direct connect 300 bps modem that is no taller than a 50c piece and fits snugly under the base of a telephone, has been released by Australian communications manufacturer, Electromed. Called the Sendata 300 the modem is simple to operate and does not require operator training. It attaches to the existing telephone wall socket plug and becomes fully operational with the flick of a switch by the operator.

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